MIL-C-46615B(MU)
7 June 1974
SUPERSEDING
MIL-C-46615A(MU)
7 April 1967

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, M29A2 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Impulse, M29A2, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS 28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Armament Command

D8593625	- Cartridge, Impulse, M29A2, Assembly
F8797306	- Box, Packing, Ammunition for Cartridge Impulse, M29A2
F8797307	- Carton, Packing, Ammunition for Cartridge Impulse, M29A2
C9225902	- Carton, Alternative, Packing, Ammunition for Cartridge, Impulse, M29A2 and M43
IEL 8593625	- Inspection Equipment List for Cartridge Impulse, M29A2

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 N2 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Vibration.-The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of any component.

3.3 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.4 Physical properties.-Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

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3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.-The cartridge when fired in an acceptable M1A3 or M4 Canopy Remover shall propel a 300 pound mass vertically and shall comply with the requirements of Table I.

TABLE I

Temperature (^{o}F)	Minimum Velocity Ft. Per Sec. (see 6.2)	Minimum Thrust-Pounds	Maximum Ignition Delay-Milliseconds (see 6.1)
-60° to -70°F	17	2000	40
65° to 75°F	20	2800	40
160° to 165°F	24	3400	40

3.5.2.1 Initiation.-The remover shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of aircraft hose, size 4, in accordance with MS 28741 at 70° $\pm$ 5°F.

3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be made in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 5 assemblies for sealant test, 30 assembled cartridges for ballistics, 12 for vibration (to be selected from ballistic sample), and 10 sets of unassembled components for visual and dimensional examination.

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility as specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Table I and Table II.

TABLE II

Temperature	Not Vibrated	Vibrated
-65° to -70°F	6	4
65° to 75°F	6	4
160° to 165°F	6	4

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

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4.3.1.2 Size of lot.-An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Primers from not more than one lot.

d. Igniters from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual characteristic the criteria in Table III shall be used. For total characteristics within a major or minor class the criteria in Table IV shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens as prescribed in ANSI B46.1 N2 - Surface Texture shall be used as a basis for comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE IV
Total Allowable Defects Per Sample Size

Sample Size for Major Defects - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

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TABLE IV (Cont'd)

Sample Size for Minor Defects - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Sealing Disc, Dwg. C8593494, covering a detail of Dwg. D8593625.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Marking missing, incorrect or illegible	Visual
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Thickness	Gage
203. Varnish missing	Visual

4.3.2.2 Firing Plug, Dwg. 8593498 covering a detail of Dwg. D8593625.

CRITICAL	
1. Length of striker tip, min.	Gage
MAJOR	
101. Diameter of body, min.	Gage
102. Profile of striker tip	Gage
103. Height of boss	Gage

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4.3.2.2 (Cont'd)

	Defect	Method of Inspection
104.	Metal defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

MINOR

201.	Diameter of body, max.	Gage
202.	Diameter of boss	Gage
203.	Total length	Gage
204.	Length of striker tip, max.	Gage
205.	Surface finish improper	Visual

4.3.2.3 Retaining Ring, Dwg. B8593499, covering a detail of Dwg. D8593625.

CRITICAL

None defined

MAJOR

101.	Concentricity of center hole with body	Gage
102.	Diameter of body, min.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

MINOR

201.	Diameter of center hole	Gage
202.	Diameter of body, max.	Gage
203.	Thickness	Gage

4.3.2.4 Cartridge Case, Dwg. C8593493, covering a detail of Dwg. D8593625.

CRITICAL

None defined

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4.3.2.4 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Concentricity of head with body	Gage
102.	Straightness of base	Gage
103.	Diameter of body, min.	Gage
104.	Diameter of head, min.	Gage
105.	Thickness of sidewall, min.	Gage
106.	Thickness of base, min.	Gage
107.	Metal defective	Visual
108.	Protective finish inadequate or defective	Visual
109.	Burr	Visual
110.	Foreign matter	Visual
MINOR		
201.	Diameter of body, max.	Gage
202.	Diameter of head, max.	Gage
203.	Total length	Gage
204.	Thickness of sidewall, max.	Gage
205.	Location of datum diameter	Gage
206.	Radius or chamfer missing or incorrect	Visual
207.	Surface finish improper	Visual

4.3.2.5 Cartridge Head, Dwg. C8593628, covering a detail of Dwg. D8593625.

CRITICAL		
None defined		
MAJOR		
101.	Concentricity of primer pocket recess with primer pocket	Gage
102.	Concentricity of primer pocket with body	Gage
103.	Concentricity of recess on top of head with primer pocket	Gage
104.	Concentricity of flange on head with body	Gage
105.	Diameter of primer pocket	Gage
106.	Diameter of primer pocket recess, min.	Gage
107.	Diameter of body, min.	Gage
108.	Diameter of flange, min.	Gage

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
109. Depth to bottom of primer pocket recess	Gage
110. Depth of primer pocket to chamfer at base	Gage
111. Location of datum diameter	Gage
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Surface finish of primer pocket, primer pocket recess and top of head	Visual
115. Burr	Visual
116. Foreign matter	Visual

MINOR

201. Diameter of primer pocket recess, max.	Gage
202. Diameter of recess on top of head	Gage
203. Diameter of large cavity bottom	Gage
204. Diameter of body, max.	Gage
205. Diameter of flange, max.	Gage
206. Depth of recess on top of head	Gage
207. Total length	Gage
208. Radius or chamfer missing or incorrect	Visual
209. Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Primer, Percussion, Dwg. D10551686, covering a detail of Dwg. D8593625.

CRITICAL

1. Metal defects	Visual
2. Primer foil discolored	Visual
3. Mixed primers	Visual

MAJOR

101. Burr	Visual
102. Foreign matter	Visual

MINOR

None defined

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4.3.2.7 Cartridge, Impulse, M29A2 Assembly Drawing D8593625.

Defect	Method of Inspection
CRITICAL	
1. Primer not properly seated	X-ray 1/ Visual
2. Any part missing	X-ray 1/ Visual
3. Improper assembly	X-ray 1/ Visual
4. Metal defects	X-ray 1/ Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Crimp missing or improper	Visual
103. Sealant missing	Visual
104. Sealant in cavity	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Diameter of head, max.	Gage
203. Total length, max.	Gage
204. Location of datum diameter	Gage
205. Varnish missing or inadequate	Visual

1/ The cartridge shall be x-rayed with the sides of the case perpendicular to the rays. The x-rayed negatives, properly identified shall be forwarded to the procuring activity.

4.3.2.8 Carton, Packing, Ammunition for Cartridge Impulse M29A2, Drawing F8797307.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
204. Heat sealer imcomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incomplete or illegible	Visual

4.3.2.9 Box, Packing, Ammunition for Cartridge, Impulse M29A2, Drawing F8797306.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.10 Carton, Alternative, Packing Drawing C9225902.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents	Manual and Visual
202. Carton broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect, or illegible	Visual
205. Tape incomplete	Visual

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Vibration.-Twelve of the 30 cartridges of the initial production sample shall be subjected to vibration. The 12 cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirements of 3.2 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.2 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of acceptability of coated parts submitted by the contractor shall be in accordance with provisions set forth in Tables III and IV and MIL-C-60539.

4.3.3.3 Hardness.-Ten units from each heat-treated lot of components shall be tested. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the lot of components.

4.3.3.4 Airtightness.-Each cartridge shall be subjected to the specified pressure differential. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.5 Sealant application.-Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.-The following test shall be performed by the Government:

4.3.3.6.1 Ballistics.-Thirty cartridges for the initial production sample, and 30 cartridge from each production lot, shall be grouped and temperature conditioned as prescribed in Table V and 4.4.6.1.1 and then tested in accordance with 4.4.6. Failure of any cartridge to comply with the requirements of 3.5 shall be cause for lot rejection.

TABLE V

Temperature	Initial Production Sample	Production Lot Sample
-65°F to -70°F	10	10
65°F to 75°F	10	10
160°F to 165°F	10	10

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4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8593625.

4.4 Test methods and procedures.

4.4.1 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.2 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and cartridge head shall be sectioned longitudinally to better expose the significant surface to the salt spray.

4.4.3 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.4 Airtightness.-The method for determining airtightness shall be by use of equipment listed on IEL 8593625.

4.4.5 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant cavity is free of sealant.

4.4.6.1 Ballistics.-The cartridges shall be loaded into acceptable M1A3 or M4 Removers, divided into 3 equal groups and temperature conditioned in accordance with Table V. The cartridge shall be tested within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the remover shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-A standard M1A3 or M4 Remover loaded with a dummy M29A2 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity.

Level A - Drawings F8797306 and F8797307
Level B - Drawings F8797307 and C9225902.

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6. NOTES

6.1 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.2 The velocity referred to shall be that of the projected mass at the moment of separation of the tubes.

6.3 Ordering data.-Procurement documents shall specify the title, number, and date of this specification.

Custodian:

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Preparing Activity:

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-46615B(MU) Cartridge, Impulse, M29A2 Assembly

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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